

Input module

Hardware specification

PEs, buffers, links, DRAM
Structure, precision, format

Scheduling scheme

Dataflow & mapping

DNN workload / PyTorch

Layer dimensions
Tensor values (optional)

Accelergy / CACTI

Main module

Simulation engine

Component models

Instantiate & connect
PE / buffer / link / DRAM

Cycle-level timing

Dependencies / contention
Buffering / overlap

Functional execution

Actual tensor arithmetic

Activity / cost accounting

Execution schedulers

Per-boundary tile schedules
Tile offsets & reuse counts
Transfer & compute order

What moves, and in what order

Tensor values (optional)

Component cost model

Output module

Runtime statistics

Execution cycles
Energy
PE utilization
Buffer occupancy
Operation count
OPS / intensity

DNN output

Inference accuracy
(functional mode)

Layers

Tiles / order

Requests

Unit costs